Design, Synthesis and Test of Reversible Circuits for Emerging Nanotechnologies

Himanshu Thapliyal and Nagarajan Ranganathan*
Department of CSE, University of South Florida, Tampa, FL, USA
Email:{hthapliy,ranganat}@cse.usf.edu; *Ph.D. Advisor

Reversible circuits are those circuits that do not lose information during computation. These circuits can generate unique output vector from each input vector, and vice versa, that is, there is a one-to-one mapping between the input and the output vectors. Among the emerging computing paradigms, reversible logic appears to be promising due to its wide applications in emerging technologies such as quantum computing, quantum dot cellular automata, optical computing, etc. Reversible logic can be useful to design non-dissipative circuits if the physical implementation platform is also physically reversible. Therefore, reversible logic is being investigated for its promising applications in power-efficient nanocomputing. Further, a quantum computer must be built from reversible logic components. Thus, the feasibility of reversible logic circuits could critically impact the realization of quantum computing.

I. Motivation

Several important metrics need to be considered in the design of reversible circuits. The constant input in the reversible quantum circuit is called the ancilla input, while the garbage output refers to the output which exists in the circuit just to maintain one-to-one mapping but is not a primary or a useful output. Quantum computers of many qubits are extremely difficult to realize thus the number of ancilla inputs and the garbage outputs in the quantum circuits needs to be minimized. The reversible circuit has other important parameters of quantum cost and delay which need to be optimized. The quantum cost of a design is the number of 1x1 and 2x2 reversible gates used in its design, thus can be considered equivalent to number of transistors needed in a conventional CMOS design. The delay of a reversible circuit can be computed by calculating its logical depth when it is designed from smaller 1x1 and 2x2 reversible gates. A synthesis framework in which a single parameter is optimized is inadequate since optimizing one parameter often results in the degradation of other important parameters. Additionally, there are number of implementation platforms such as trapped ions, spintronics, superconducting circuits, that are being explored for physical implementation of qubits and quantum gates. Currently, it is not sure which implementation technology will be the future of the quantum computers. *Thus there is a need of technology independent design and synthesis of reversible logic circuits that are applicable to quantum computing*. Further, nanoelectronic transistors tend to be highly sensitive to environmental influences, such as temperature, cosmic ray particles and background noise-producing transient faults. Thus, there is need of advance novel techniques and design methodologies that can address the online (concurrent) as well as offline testing of faults in circuits based on emerging nanotechnologies.

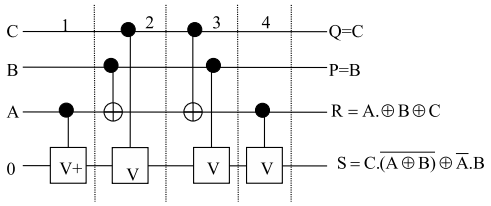

Fig. 1. Delay optimized design of TR gate based reversible full subtractor

II. Contributions of Dissertation [1]

The first contribution of this dissertation is the design of a new reversible gate namely the TR gate (Thapliyal-Ranganathan) which has unique structure that makes it ideal for the realization of arithmetic circuits such as adders, subtractors and comparators, efficient in terms of the parameters such as ancilla and garbage bits, quantum cost and delay [2], [3]. An example of delay optimized design of TR gate based 1 bit full subtractor having quantum cost of 6 and delay of 4 Δ is shown in Fig. 1 [1].

The second contribution is the development of design methodologies and a synthesis framework to synthesize reversible data path functional units, such as binary and BCD adders, subtractors, adder-subtractors and binary comparators. The objective behind the proposed design methodologies is to synthesize arithmetic and logic functional units optimizing key metrics such as ancilla inputs, garbage outputs, quantum cost and delay. A library of reversible gates such as the Fredkin gate, the Toffoli gate, the TR gate, etc. was developed by coding in Verilog HDL for use during synthesis [4], [5], [6], [7]. We propose to integrate the various design methodologies for arithmetic and logic units as a software tool suite as shown in Figure 2. The synthesis portion consists of the algorithms that embed the proposed design methodologies and do not follow the traditional approach due to the variability in the various design methodologies covering a wide range of arithmetic and logic circuits. The proposed framework consists

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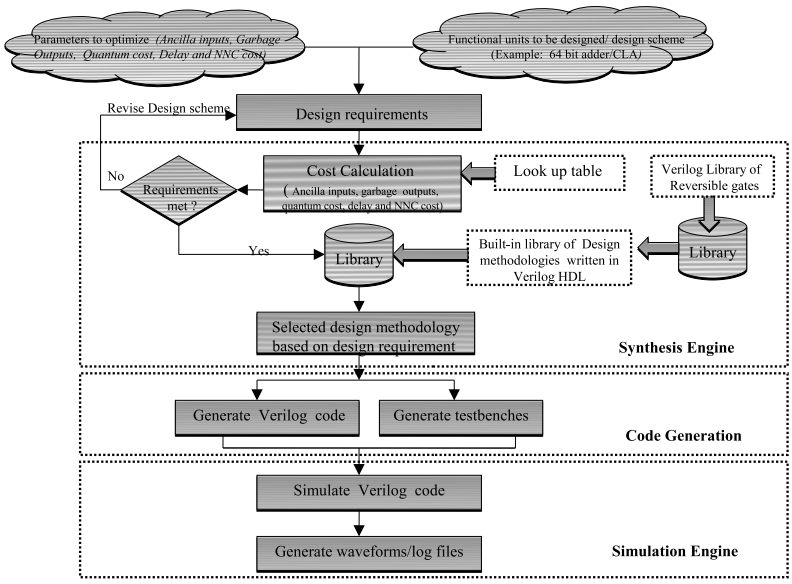

Fig. 2. Proposed synthesis framework for design of reversible arithmetic and logic circuits

of three main components: synthesis engine, code generation and the simulation engine [1].

The third contribution is the set of methodologies for the design of reversible sequential circuits such as reversible latches, flip-flops and shift registers. The reversible designs of asynchronous set/reset D latch and the D flip-flop are attempted for the first time. It is shown that the designs are optimal in terms of number of garbage outputs while exploring the best possible values for quantum cost and delay [8], [9].

The other important contributions of this dissertation are the applications of reversible logic as well as a special class of reversible logic called conservative reversible logic towards online (concurrent) and offline testing of single as well as multiple faults in traditional and reversible nanoscale VLSI circuits, based on emerging nanotechnologies such as QCA, quantum computing, etc. Nanoelectronic devices tend to have high permanent and transient faults and thus are susceptible to high error rates. Specific contributions include (i) concurrently testable sequential circuits for molecular QCA based on reversible logic [10], (ii) concurrently testable QCA-based FPGA [11], (iii) design of self checking conservative logic gates for QCA [12], [13], [14], (iv) concurrent multiple error detection in emerging nanotechnologies using reversible logic [15], (v) two-vectors, all 0s and all 1s, testable reversible sequential circuits [16], [17].

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